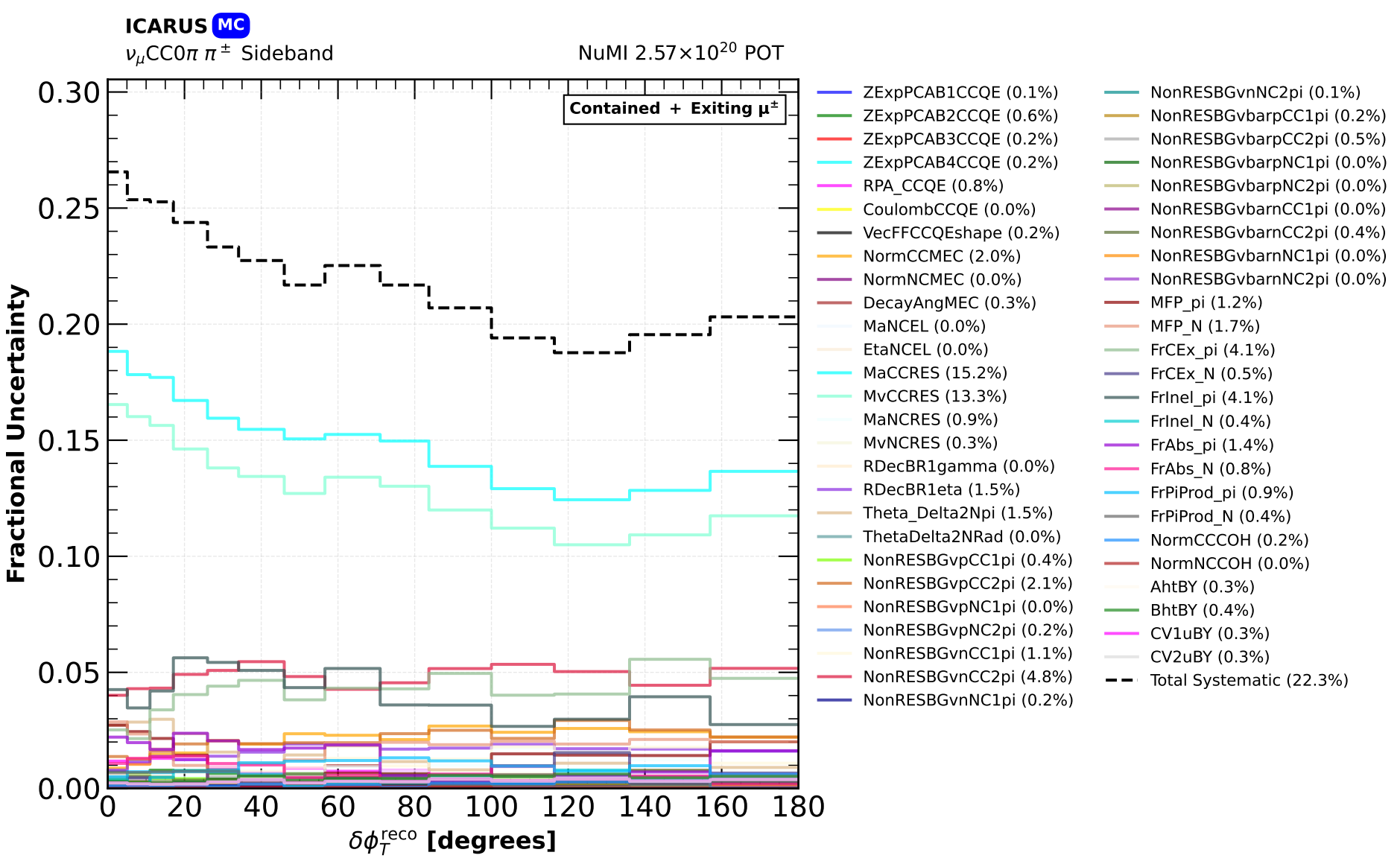
ICARUS MC
ν_μCC0π $\pi^\pm$ Sideband
NuMI 2.57×10^{20} POT
Contained + Exiting μ$^\pm$
Fractional Uncertainty
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
20
40
60
80
100
120
140
160
180
$\delta\phi_T^{\text{reco}}$ [degrees]
ZExpPCAB1CCQE (0.1%)
ZExpPCAB2CCQE (0.6%)
ZExpPCAB3CCQE (0.2%)
ZExpPCAB4CCQE (0.2%)
RPA_CCQE (0.8%)
CoulombCCQE (0.0%)
VecFFCCQEshape (0.2%)
NormCCMEC (2.0%)
NormNCMEC (0.0%)
DecayAngMEC (0.3%)
MaNCEL (0.0%)
EtaNCEL (0.0%)
MaCCRES (15.2%)
MvCCRES (13.3%)
MaNCRES (0.9%)
MvNCRES (0.3%)
RDecBR1gamma (0.0%)
RDecBR1eta (1.5%)
Theta_Delta2Npi (1.5%)
ThetaDelta2NRad (0.0%)
NonRESBGvpCC1pi (0.4%)
NonRESBGvpCC2pi (2.1%)
NonRESBGvpNC1pi (0.0%)
NonRESBGvpNC2pi (0.2%)
NonRESBGvnCC1pi (1.1%)
NonRESBGvnCC2pi (4.8%)
NonRESBGvnNC1pi (0.2%)
NonRESBGvnNC2pi (0.1%)
NonRESBGvbarpCC1pi (0.2%)
NonRESBGvbarpCC2pi (0.5%)
NonRESBGvbarpNC1pi (0.0%)
NonRESBGvbarpNC2pi (0.0%)
NonRESBGvbarnCC1pi (0.0%)
NonRESBGvbarnCC2pi (0.4%)
NonRESBGvbarnNC1pi (0.0%)
NonRESBGvbarnNC2pi (0.0%)
MFP_pi (1.2%)
MFP_N (1.7%)
FrCEx_pi (4.1%)
FrCEx_N (0.5%)
FrInel_pi (4.1%)
FrInel_N (0.4%)
FrAbs_pi (1.4%)
FrAbs_N (0.8%)
FrPiProd_pi (0.9%)
FrPiProd_N (0.4%)
NormCCCOH (0.2%)
NormNCCOH (0.0%)
AhtBY (0.3%)
BhtBY (0.4%)
CV1uBY (0.3%)
CV2uBY (0.3%)
Total Systematic (22.3%)